

UNIT I

Programming Languages

What is a computer?

A computer is an electronic device that manipulates information, or data. It has the ability to store, retrieve, and process data. You may already know that you can use a computer to type documents, send email, play games, and browse the Web. You can also use it to edit or create spreadsheets, presentations, and even videos.

Basically A computer system is divided into two categories.

1. Hardware
2. Software

Software

Software is a set of instructions, data, or programs used to operate a computer and execute specific tasks. In simpler terms, software tells a computer how to function. It's a generic term used to refer to applications, scripts, and programs that run on devices such as PCs, mobile phones, tablets, and other smart devices. Software contrasts with hardware, which is the physical aspects of a computer that perform the work.

There are three types of Software used in computer System.

1. System Software
 - a. DOS, Windows, Linux, Unix, Ubuntu
2. Application Software
 - a. Word, Excel, Powerpoint, Access, Notepad.
3. Programming Languages.
 - a. C, C++, JAVA, .NET, Python, R-Language

What is language?

A language means a medium of communication from one person to another. It is exchange of facts ideas, opinions or emotions by two or more person. This language needed for carrying out the day to day activities of human life.

Marathi, Hindi, English etc are example of natural language used to communicate the people. Each language has its own set of alphabets, words to make meaningful instructions. All these natural language follows rules of spelling and grammar.

Similarly to communicate with computer for specific kind of work, which should be known by the computer system, we need the language called as programming language. This language is used for proper functioning of system; perform various operations such as arithmetical, logical according to the instructions given in the form of computer language.

What is programming language?

The computer does not understand English so in order to interact with computer or to make a communication with the computer the programmer must know the language which the computer understands. The information and instruction which are to be carried out by the computer are written in the form of a program.

Programming language is a medium to communicate with a computer. The programming language consists of alphabets, symbols, words & sentences. The programming language written in such a way that the computer reads, understands and interprets each statement finally gives desired result according to the manner of program instructions.

Types of languages

1.1. Machine Language

1.2. Assembly language

1.3. High level languages

'C' is a general purpose, Structured Programming language. It is suitable for solving scientific and engineering problems as well as for business application. Over the past few years, C has become the most popular computer programming language. C's speed and power have always made it a favorite language for operating systems, compilers, interpreters and word processors. Many powerful applications can be generated in C language economically and correctly using C language. C language encourages users to write additional library functions of their own to enhance the features of C language. One of the unique features of C language is that it is available on any make of computer and also available for a range of computers, including mainframe, mini and microcomputers.

Types of languages

As we human beings communicate with each other's in different language such as English, French, German or Arabic. Similarly to communicate with the computers we have to use specific language and for this purpose hundreds of languages have been invented in which few of them has gained international reputation. C language is one of them. Languages can be broadly categorized into three categories.

Machine Language (Low Level Language)

The most elementary and first type of computer, which was invented, was machine language. Machine language was machine dependent. A program written in machine language cannot be run on another type of computer without significant alterations. Machine language is sometimes referred to as the binary language; i-e, the language of 0 and 1, where 0 stands for the absence of electric pulse and 1 stands for the presence of electric pulse. Very few computer programs are actually written in machine language.

Assembly language

As computer became more popular, it became quite apparent that machine language programming was simply too slow and tedious for most programmers. Assembly languages are also

called as low-level language. Instead of using the string of numbers, programmers began using English-like abbreviations to represent the elementary operation. The language provided an opportunity to the programmers to use English like words that were called MNEMONICS.

For Example

LDA A

ADD B

Where

LDAA = Load Accumulator with A

Add B = Add number B to the contents of the accumulator, leaving the result in accumulator.

High level languages

The assembly languages started using English like words, but still it was difficult to learn these languages. High level languages are the computers language in which it is much easier to write a program than the low-level languages. A program written in high-level language is just like giving instruction to person in daily life. It was in 1957 that a high level language called FORTRAN (Formula Translation) was developed by IBM, which was specially developed for Scientists and Engineers. Other high level languages are COBOL (Common Business Oriented Language), which is widely used for business data processing task. BASIC (Beginners All-Purpose Symbolic Instruction Codes), which is developed for the beginners in 1960s. The most powerful and widely used language is C language which is a general purpose programming language. You can use C language for almost any programming task. PASCAL PL/1 are other high level languages, which has gained widespread acceptance.

Compiler and Interpreter

We generally write a computer program using a high-level language. A high-level language is one which is understandable by us humans. It contains words and phrases from the English (or other) language. But a computer does not understand high-level language. It only understands program written in 0's and 1's in binary, called the machine code. A program written in high-level language is called a source code. We need to convert the source code into machine code and this is accomplished by compilers and interpreters. Hence, a compiler or an interpreter is a program that converts program written in high-level language into machine code understood by the computer.

The difference between an interpreter and a compiler is given below:

Compiler

Scans the entire program and translates it as a whole into machine code. It takes large amount of time to analyze the source code but the overall execution time is comparatively faster. Generates intermediate object code which further requires linking, hence requires more memory. It generates the error message only after scanning the whole program. Hence debugging is comparatively hard. Programming language like C, C++ use compilers.



Interpreter

Translates program one statement at a time. It takes less amount of time to analyze the source code but the overall execution time is slower. No intermediate object code is generated, hence are memory efficient. Continues translating the program until the first error is met, in which case it stops. Hence debugging is easy. Programming language like Python, Ruby use interpreters.



Figure: Interpreter